



Actiquest Inc. *Partnership Proposal*

AI Engineering Partner for EPC Digitalization



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CEO, Founder

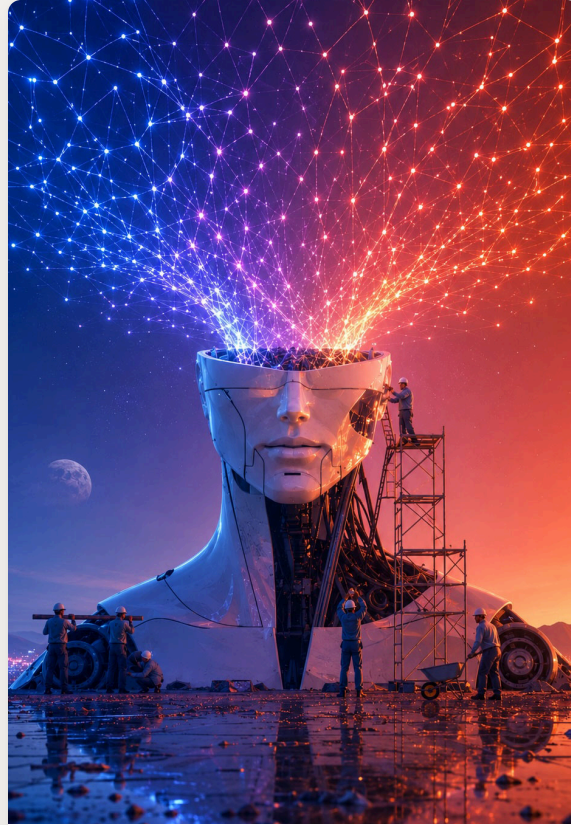
Who We Are



Actquest Labs

Applied AI Systems Company

Actquest Labs builds applied AI systems for companies that need practical automation, intelligent workflows and enterprise-grade AI products.



Membria.AI

AI Operating Environment

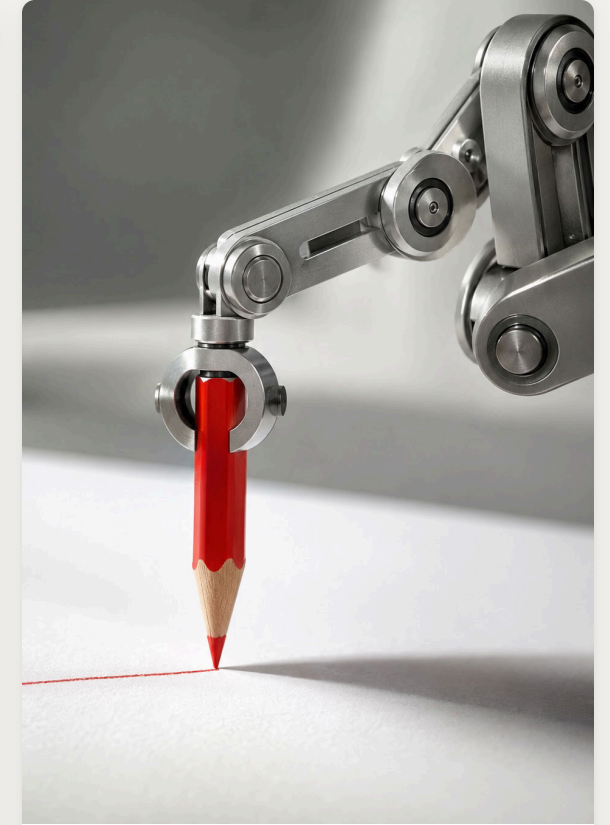
Our core product — agents, memory, knowledge and verification combined into one enterprise-ready AI platform.



Vertical Expertise

Code · Fintech · Government · Audit · EPC

We specialize in complex domains where AI needs precision, context and process understanding — fintech, government, audit, compliance and EPC.



How We Work

Venture Studio + Enterprise Delivery

We build our own AI products, partner with startups and deliver enterprise AI projects, from concept and prototype to production deployment.

Actquest Labs at a Glance

Key facts, production footprint and enterprise AI capabilities.

10+

years enterprise product, AI engineering and venture-building experience

4

verticals
Fintech · Government · Audit · Engineering / EPC

100+

seats deployed across production AI use cases

Metric	Value
Founded	February 2023 · California, USA
Company Focus	Applied AI systems · Product development · Enterprise AI
Core Product	Membria.AI — AI Operating Environment
Production Verticals	SDLC · Fintech / Banking · AI Accounting / Audit · Engineering / EPC
Deployment	Cloud · On-premise · Enterprise environments
Team	AI engineers · Product managers · Domain experts · Venture builders
Production Use	100+ deployed seats across active use cases
Delivery Model	Venture studio · Enterprise projects · Strategic partnerships
Clients & Partners	OES · Wowcube · Hiscox · K-Telco · Kazakhstan Government · Large industrial enterprises across EMEA / CIS
Technology Partners	NVIDIA · Amazon Web Services
Team Background	20+ years in enterprise software and audit · 9 years in applied AI — across financial services, government, industrial and deep-tech sectors

From AI concept to production-grade enterprise systems.

Membria Reasoning Graph Workforce

INSTITUTIONAL KNOWLEDGE



Memory

- ✓ **Decisions** — every choice captured with its reasoning
- ✓ **Consistency** — behavior chains keep teams aligned
- ✓ **Skills** — wins distilled into reusable know-how
- ✓ **NegativeKnowledge** — failures become «do-not» rules

MULTI-AGENT HARNESS



Reasoning

- ✓ **Mesh** — many agents work in parallel
- ✓ **Coordination** — agents hand off via A2A
- ✓ **Routing** — each task to the right model
- ✓ **Quality gates** — verification blocks bad output

DOMAIN-AGNOSTIC FOUNDATION



Graph

- ✓ **GraphRAG** — semantic search over a property graph
- ✓ **Multi-domain** — one schema across every domain
- ✓ **Orchestration** — work runs as a dependency graph
- ✓ **Adaptive** — memory that strengthens with use

Proven Across 4 Domains

One AI substrate · any vertical domain



Software Development / AI-PDLC

Senior engineers leave and take their judgment. Membria keeps every decision — teams stop re-solving the same problems.

AI LAYER
Code knowledge graph · decision memory · behavior guidance

VALUE
Fewer regressions · faster onboarding · knowledge that stays



Banking / Finance

A new regulation meant a six-week release. Now it's live on the next transaction — every posting traceable to the clause that authorized it.

AI LAYER
Policy graph · automated ledger entries · live rule updates

VALUE
Full traceability · no release cycle · operationally instant



Glass — AI Accounting/Audit

Audit teams lose weeks tracing entries to source documents. Membria marks every posting with origin and trust score — unsupported entries surface on day one.

AI LAYER
Source-linked entries · trust scoring · automatic gap detection

VALUE
Audit-ready by default · transparent · no retrofitting



AI-EPC / Engineering

A missed P&ID clause becomes a safety incident. Five AI critics check every diagram against 50+ standards before any engineer opens the file.

AI LAYER
Process · Mechanical · HSE · I&C · Piping — automated risk scoring

VALUE
Faster reviews · fewer missed risks · consistent across disciplines

Common pattern across domains: domain knowledge → graph memory → AI workflows → human verification → production deployment

Core Capabilities

The full stack — from memory and reasoning to deployment and governance



AI Agent Orchestration

Multi-agent BBS mesh, WorkGraph DAG, A2A protocol coordination



Causal Memory

Decision / Outcome / NegativeKnowledge graph with time-decay and compounding



Verification Engine

Cross-document consistency checks, rule-based normative validation, quality gates



Graph Intelligence

FalkorDB property graph, symbol-level CodeGraph, policy graph with Hebbian co-activation



LLM Integration

Claude / GPT factory, structured output, behavior chains, context injection



Product Engineering

Full-stack: FastAPI + React + Node.js, MCP protocol, Telegram, Web Dashboard



Domain AI Systems

Vertical-specific schemas, critics, standards compliance (ISA / API / ASME / IEC)



Security & Governance

PolicyEngine, RBAC, OIDC/SSO, prompt injection defense, secrets management, rate limiting

Bottom line: Generic AI forgets. Membria remembers, reasons, and governs.

AI-CAD for *Engineering* (AI-EPC)

Production-grade engineering AI — built to be trusted on safety-critical drawings



Production-grade P&ID output

ISA 5.1 symbols exported to DEXPI XML, DXF and IFC4 — drawings that drop straight into your existing pipeline.



Works with the tools you already run

DEXPI interchange with AVEVA and SmartPlant — no rip-and-replace of your toolchain.



Deterministic, not generative

A rule engine — not a guessing model. No hallucinations on safety-critical drawings.



Five-discipline AI review

Process, Mechanical, HSE, I&C and Piping critics catch errors before sign-off.



Safety analysis built in

HAZOP, SIL matrix and Cause-&-Effect generated automatically — not bolted on after.



Defensible audit trail

Every decision recorded with the alternatives considered and the reasoning behind it.

Current EPC AI — Nobody Remembers

Every EPC AI tool today is stateless — each request starts from zero. None offer persistent agent memory for engineering projects.

Player	What they do	Persistent memory?	Why not a direct competitor
Autodesk	Design + construction cloud	No	AI = copilot for CAD, not knowledge management
Bentley Systems	Infrastructure digital twins	No	Twin ≠ memory — no decision provenance, no strategy extraction
Trimble	"Agent Studio" pilot for E&C	No	Early pilot, no persistent memory across sessions
Procore	Helix AI automation + Datagrid	No	Stateless AI — every request starts from zero, no cross-session memory
AVEVA / Schneider	Process industry digital twins	No	Operational data, not engineering decisions
SymphonyAI	P&ID intelligence	No	Drawing recognition, not a memory plane

The real competition is inertia: Excel + "it works fine as is." Every project starts from a blank page — 28% of time lost to rework, +28% budget overrun on average. That is the memory problem.

The EPC Whitespace

Engineering companies don't need another document chatbot. They need project memory that survives handovers, reviews and future projects.

EPC pain	Current problem	Membria response
Cross-phase handover	FEED assumptions disappear before detail design and construction	Cross-phase memory persistence
Scope changes	Decisions scattered across emails, PDFs, CAD, meetings and issue logs	Scope assumption capture with full provenance
Rework	Teams repeat mistakes because lessons learned are not operationalized	Negative knowledge and reusable strategies
Engineering review	Generic AI cannot verify discipline-specific logic	Human + AI verification workflows (5 critics)
Procurement memory	Supplier, schedule and specification lessons are lost project to project	Procurement/vendor memory graph
Compliance	Audit trails incomplete or hard to reconstruct before handover	Immutable mutation ledger and compliance packs

Market signal: 9 of 10 large EPC projects exceed budget (+28% avg) · 18% of time lost searching for data · 28% lost to rework. This is the memory problem — and it's solvable.

Where We Win — and Where We Must Prove Scale

WHERE MEMBRIA WINS

✓ Memory lifecycle

candidate → validated → consolidated → strategy. 8 states, maturity scoring, automatic promotion — 7 memory types vs 4 in Palantir, 3 in Microsoft.

✓ Policy-aware retrieval

relevance × utility × freshness × permission × risk × budget. Safety engineers see only safety facts — automatic, not manual filtering.

✓ External memory plane API

Any agent — yours or third-party — reads and writes policy-aware context through one open API. Palantir's Ontology is closed.

✓ EPC-native use cases

P&ID memory, cross-phase handover, procurement sequencing, engineering decisions with provenance — built in, not bolted on.

✓ Compliance-grade audit

Immutable mutation ledger, deletion certificates, decision packs — EU AI Act and ISO 19650 ready out of the box.

✓ Negative knowledge

Explicit graph nodes for what should not be repeated — with evidence, context and project history. No competitor does this.

WHAT WE MUST PROVE

→ Scale

1000+ users, large enterprise datasets, load testing at production volume — architecture ready, load testing in progress.

→ Platform integrations

AVEVA, Bentley, SmartPlant, Autodesk, EDMS and ERP connectors — on the roadmap for Phase 2.

→ Certifications

FedRAMP / ISO 27001 / enterprise security audit path — planned alongside Phase 2 deployments.

→ Visual engineering intelligence

P&ID image recognition and CAD/BIM extraction for brownfield legacy drawings — early R&D, partner-able capability.

→ Enterprise community

Developer templates, marketplace, partner ecosystem — building with early design partners.

Why Now

Large EPC companies run **multi-billion dollar projects** — and the knowledge needed to execute them is scattered across documents, disciplines and rotating teams.

An engineer re-solves a P&ID problem solved two projects ago. A compliance clause slips past a busy reviewer. A senior engineer leaves and takes fifteen years of judgment with them.

01

EPC is digitizing — but knowledge stays siloed

Complex EPC projects across **oil & gas, petrochemical and industrial sectors** each generate thousands of drawings, specifications and decisions that remain isolated per project.

→ The knowledge already exists — it just isn't connected

02

The missing layer is engineering intelligence

Each discipline has its tools. The opportunity is a governed layer that **connects P&ID verification, standards compliance and cross-phase project knowledge** — not just within a project, but across all of them.

→ Connected intelligence reduces rework and speeds delivery

03

The moment is now

EU AI Act and ISO 19650 revision create **demand for traceable decisions and immutable audit trails** in safety-critical infrastructure — exactly what Membria already delivers.

→ Regulation is creating demand for what we already built

Where We Can Add *Value*

One Membria substrate, two fronts — the **engineering AI layer** for project execution and design verification, the **knowledge layer** for institutional continuity across projects.

ENGINEERING & DESIGN



P&ID AI

AI-CAD generating and verifying P&IDs against ISA 5.1 / API / ASME standards — five-discipline critics before any sign-off.



Safety Analysis

HAZOP, SIL matrix and Cause-&-Effect generated automatically from the diagram — not bolted on after review.



Document Intelligence

Semantic search over project documentation, specs and datasheets — engineers find answers in seconds, not hours.



Design Verification

Cross-document consistency checks catch conflicts between P&IDs, datasheets and specs before they reach the field.

PROJECT & INSTITUTIONAL



Project Memory

Every design decision captured with its reasoning — survives team rotation on multi-year projects and carries forward to the next.



Standards Compliance

Automatic normative validation against ISA, API, ASME, IEC — compliance by default, not by firefighting before handover.



Audit Trail

Every change traceable to the clause that authorized it — punch-list and handover package ready from day one, not assembled at the end.



Procurement AI

Smart recommendations based on historical project data, vendor performance and specifications — negative knowledge that compounds across projects.

Cross-phase persistence: Membria indexes every drawing, spec and decision into a graph — project knowledge that doesn't walk out the door on handover day.

Why *Actiquest Labs*?



AI-Native from Ground Up

Built architecture-first for multi-agent, memory-driven AI systems from day one, not retrofitted onto legacy software.



Proven Flexibility

One platform, four production verticals: software dev, finance, accounting, industrial engineering. Domain-agnostic by design.



Memory Compounds

Our moat is causal memory — institutional knowledge that grows with usage and survives team turnover.



Verification First

We don't just generate — we verify. Consistency checks, normative compliance, and quality gates before output.



Full-Stack Execution

Strategy + engineering + product + launch in one team. No handoff friction between thinking and building.



Speed and Rigor

We move fast to prototype but maintain audit trails, policy governance, and enterprise-grade security from day one.

What this means for you: a working prototype in weeks · a full AI/product team without the hiring overhead · validation before any major investment.

Collaboration Options

A graduated on-ramp — **start small, scale as the value proves out.** No need to commit to a venture on day one.

01

RECOMMENDED START

Pilot Project

Goal: Validate one EPC AI use case end-to-end.

- Working prototype
- Demo & technical report
- Feasibility assessment

MINIMAL · FOCUSED · WEEKS

02

MVP Development

Goal: Build the first usable AI product for production.

- MVP application
- Architecture documentation
- Roadmap & test results

STRUCTURED · MONTHS

03

Innovation Partner

Goal: Continuous AI innovation support.

- Regular prototypes & research
- Product concepts
- Engineering support

LONG-TERM · ONGOING

04

Venture Co-Labs

Goal: Co-develop AI ventures and licensing for EPC.

- Joint product development
- Venture co-creation
- Technology licensing

STRATEGIC · EQUITY-LEVEL



From isolated documents to a
connected engineering brain.

Let's Work Together

One 60–90 min discovery session to define one EPC AI use case — and a pilot roadmap in your hands **within 7–10 days.**

1 Discovery Session · Week 1

Align on use case and success criteria. Understand priorities and constraints.

2 Pilot Scoping · Week 2

Technical feasibility, architecture outline, team alignment, NDA and data security.

3 Pilot Kickoff · Week 3–10

Working prototype development begins. Weekly syncs, milestone reviews, ROI tracking.

LEADERSHIP

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APPENDIX

Technology Stack

AI & LLMs

- BYOK multi-provider factory · cloud or fully local (Ollama)
- 1536-dim embeddings native in FalkorDB · local MiniLM
- 9 adaptive memory systems: Hebbian · STALE · A-MAC · FSFM · ADAM

Data & Graph

- Graph DB: FalkorDB (RedisGraph, property graph) + native vectors
- Search: Tantivy full-text · tree-sitter code graph
- Ingestion: DEXPI/Proteus · IFC · DXF · Excel · PDF (OCR) · web

Runtime & API

- Monty runtime VM (pause / resume) · A2A agent mesh · MCP (86 tools)
- Rust port → on-device agents · local model router · no cloud
- Python 3.11 · FastAPI · Node.js / Express · EventBus · pytest

Frontend & Infra

- React 18 · TypeScript 5 · Tailwind · Vite · CodeMirror · xterm.js
- Deploy: Docker / Compose · on-prem ready · 300+ CLI commands
- Auth: Keycloak OIDC · OAuth2 · RBAC · JWT · Infisical secrets